11 Double and Triple Integrals

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Introduction

Double and triple integrals can be used to evaluate areas, volumes and centres of mass, and to calculate various physical quantities, like moments of inertia.

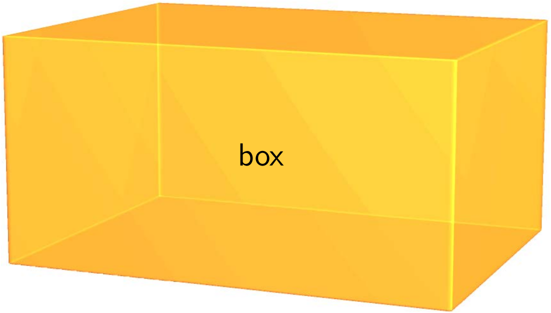
Later we will discuss path and surface integrals, along with integral theorems that can be viewed as generalisations of the fundamental theorem of calculus.

Rectangular regions

Let $D \subset \mathbb{R}^n$ be a rectangular region

$$D = \{\mathbf{x} = (x_1, \ldots, x_n) \mid a_1 \leq x_1 \leq b_1, \ldots, a_n \leq x_n \leq b_n\}$$
$$= \underline{[a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]},$$

where $a_i < b_i$ for all $i \in \{1, \ldots, n\}$.



$$\mathrm{Vol}\,(D) = (b_1 - a_1) \times (b_2 - a_2) \times \cdots \times (b_n - a_n)$$

- length, area, volume

Partition of a line segment

Let $[x_0, x_1, \ldots, x_m]$ be such that

$$a = x_0 < x_1 < x_2 < \cdots < x_m = b,$$

then the line segments $P_1, P_2, \ldots, P_m$,

$$P_i = [x_{i-1}, x_i], \quad i \in \{1, \ldots, m\}$$

form a partition of the line segment $[a, b]$.

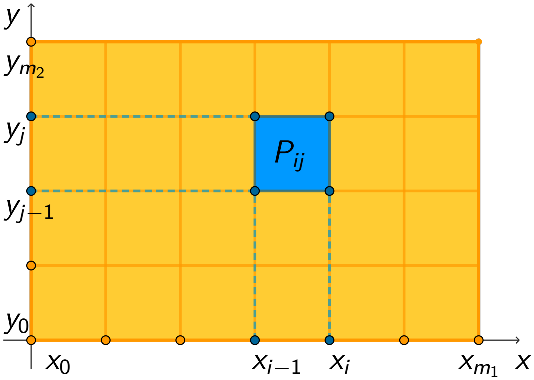
Partition of a rectangular region

If $D = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$, then, for instance, when $D \subset \mathbb{R}^2$, we have

$$a_1 = x_0 < x_1 < \cdots < x_{m_1} = b_1, \qquad a_2 = y_0 < y_1 < \cdots < y_{m_2} = b_2,$$

$$\mathcal{P} = \{P_{ij} = \underline{[x_{i-1}, x_i]} \times \underline{[y_{j-1}, y_j]} \mid i \in \{1, \ldots, m_1\},\ j \in \{1, \ldots, m_2\}\}.$$

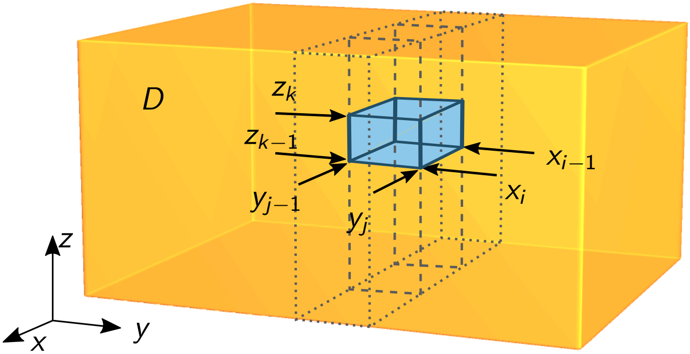
We say that $\mathcal{P}$ is a partition of D.



Partition of a rectangular region

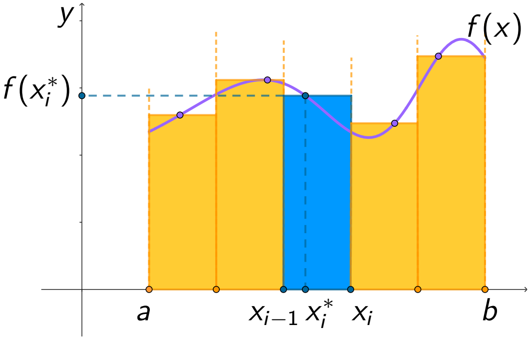
Likewise for $D = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3] \subset \mathbb{R}^3$

$$\mathcal{P} = \{[x_{i-1}, x_i] \times [y_{j-1}, y_j] \times [z_{k-1}, z_k] \mid \\ i \in \{1, \ldots, m_1\},\ j \in \{1, \ldots, m_2\},\ k \in \{1, \ldots, m_3\}\}.$$



Riemann sum

Suppose $f : [a, b] \to \mathbb{R}$ is bounded. We can approximate the area under the graph of f by calculating the **Riemann sum** of the areas of rectangles above the segments of the partition, $S^*_{\mathcal{P}}(f) = \sum_{i=1}^{m} f(x_i^*)\Delta x_i$, where where $\Delta x_i = x_i - x_{i-1}$.



The choice of $x_i^* \in [x_{i-1}, x_i]$ for $i \in \{1, \ldots, m\}$ is called the **tag** of the partition.

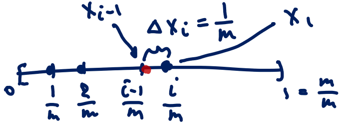
Example 11.1

Let $f(x) = x^2$, and suppose that $[a, b] = [0, 1]$. Compute the Riemann sum

$$\int_0^1 x^2\,dx = \frac{1}{3}$$

$$S^*_{\mathcal{P}}(f) = \sum_{i=1}^{m} f(x_i^*)\Delta x_i.$$

Use a uniform partition of the line segment $[0, 1]$, and choose the rightmost point of each partition segment as tag.



$$S_P^*(f) = \sum_{i=1}^{m}\left(\frac{i}{m}\right)^2\frac{1}{m} = \frac{1}{m^3}\sum_{i=1}^{m} i^2$$

$$= \frac{1}{m^{\cancel{3}\,2}}\cdot\frac{\cancel{m}(m+1)(2m+1)}{6} = \frac{2m^2+3m+1}{6m^2} \xrightarrow[m\to\infty]{} \frac{1}{3}$$

Calculating a Riemann sum (continued)

Riemann sum in higher dimensions

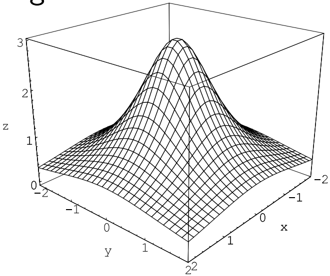
For $D = [a_1, b_1] \times [a_2, b_2]$ we consider the Riemann sum

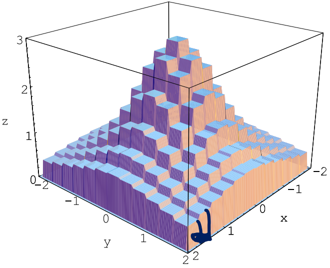
$$S^*_{\mathcal{P}}(f) = \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \underbrace{f(\mathbf{x}^*_{ij})}\underbrace{\Delta x_i \Delta y_j},$$

where $\mathbf{x}^*_{ij}$ is the tag. Same for $D = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$,

$$S^*_{\mathcal{P}}(f) = \underbrace{\sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \sum_{k=1}^{m_3} f(\mathbf{x}^*_{ijk}) \Delta x_i \Delta y_j \Delta z_k},$$

where $\mathbf{x}^*_{ijk}$ is the tag.





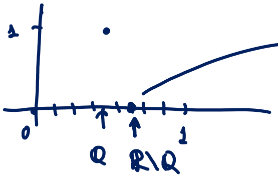
Let's do something silly

Example 11.2

Let $f : [0, 1] \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

The Riemann sums converge to different limits as partition size goes to zero, depending on the choice of tag.



depending on the choice of tag,
can get Riemann sums
converge to different quantities (0 and 1)

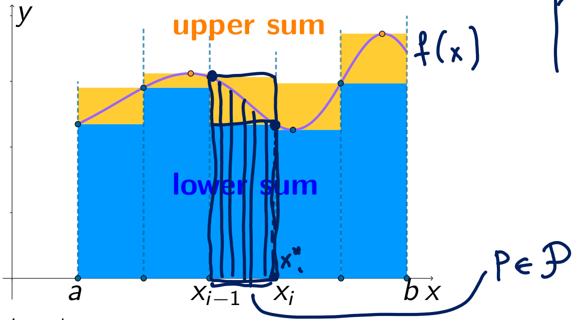
Lower and upper sums

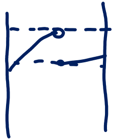
Let $D \subset \mathbb{R}^n$ be a box region, and suppose that $\mathcal{P}$ is a partition of D. We define lower and upper Darboux sums as

$$\boxed{\underline{S}_{\mathcal{P}}(f) = \sum_{P \in \mathcal{P}} \underline{f}_P \mathrm{Vol}\,(P),} \quad \overline{S}_{\mathcal{P}}(f) = \sum_{P \in \mathcal{P}} \overline{f}_P \mathrm{Vol}\,(P),$$

where

$$\underline{f}_P = \inf_{x \in P} f(x), \quad \overline{f}_P = \sup_{x \in P} f(x).$$





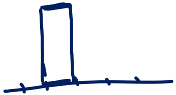
Definition of the Riemann Integral

Let D be a rectangular region, and suppose that $f : D \to \mathbb{R}$ is bounded. We say that f is (Riemann) **integrable** on D if

$\forall$ partitions $\mathcal{P}_1$ & $\mathcal{P}_2$ $\quad \underline{S}_{\mathcal{P}_1}(f) \leq \overline{S}_{\mathcal{P}_2}(f)$

$$I := \sup\{\underline{S}_{\mathcal{P}}(f) \mid \mathcal{P} \text{ partitions } D\} = \inf\{\overline{S}_{\mathcal{P}}(f) \mid \mathcal{P} \text{ partitions } D\}.$$

The value I is called the integral of f over D, and can be denoted by



$$\boxed{I = \int_D f(\mathbf{x})\, d\mathbf{x}.}$$

We can define the integral equivalently as the limit of the Riemann sum $S^*_{\mathcal{P}}(f)$ as the diameter of the partition regions goes to zero.

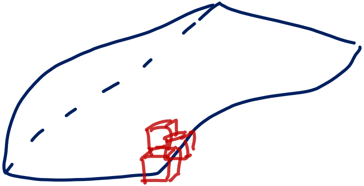
We may use a range of different ways to denote integrals throughout the course. Commonly the integral of f over D is denoted by

$$\int_D f(x)\,dx, \quad \iint_D f(x,y)\,dx\,dy, \quad \iiint_D f(x,y,z)\,dx\,dy\,dz,$$

respectively for $D \subset \mathbb{R}, \mathbb{R}^2$ and $\mathbb{R}^3$.

Sometimes it may be more convenient to use $\int_D f$, $\int_D f(\mathbf{x})\,d\mathbf{x}$, $\int_D f\,dA$ or $\int_D f\,dV$ instead.

$D \subset \mathbb{R}^2$

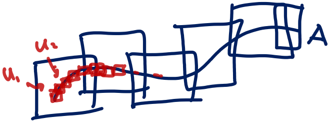


Recognising integrable functions

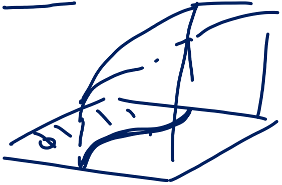
Theorem 11.3

If $D \subset \mathbb{R}^n$ is a rectangular region, and $f : D \to \mathbb{R}$ is a continuous function, then f is integrable on D.

A subset A of $\mathbb{R}^n$ has (n-dimensional) measure 0 if for any $\varepsilon > 0$ there is a cover $\{U_1, U_2, \dots\}$ of A by rectangular regions such that



$$\sum_{i=1}^{\infty} \underbrace{\mathrm{Vol}\,(U_i)}_{\text{area}(U_i)} < \varepsilon.$$



If A is compact, it is sufficient to consider finite covers.

Theorem 11.4

Let D be a rectangular region and $f : D \to \mathbb{R}$ be a bounded function. Then f is integrable if and only if the set $\{x \mid f \text{ is not continuous at } x\}$ has measure 0.

Note that graphs of continuous functions have measure 0.

Basic properties of integrals

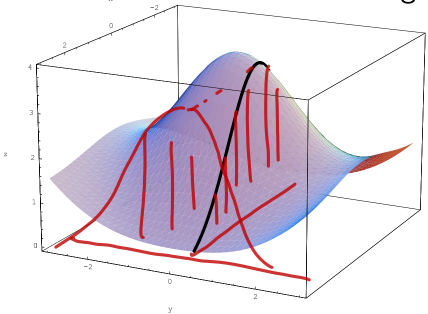
1. If $f(\mathbf{x}) \equiv 1$, $\displaystyle\int_D f(\mathbf{x})\,d\mathbf{x} = \mathrm{Vol}\,(D)$.
2. If the measure of D is zero, then $\int_D f(\mathbf{x})\,d\mathbf{x} = 0$.
3. $\displaystyle\int_D [f(\mathbf{x}) \pm g(\mathbf{x})]\,d\mathbf{x} = \int_D f(\mathbf{x})\,d\mathbf{x} \pm \int_D g(\mathbf{x})\,d\mathbf{x}$ (linearity).
4. $\displaystyle\int_D kf(\mathbf{x})\,d\mathbf{x} = k\int_D f(\mathbf{x})\,d\mathbf{x}$ for any $k \in \mathbb{R}$ (homogeneity).
5. If $f(\mathbf{x}) \geq g(\mathbf{x})$ on D then $\displaystyle\int_D f(\mathbf{x})\,d\mathbf{x} \geq \int_D g(\mathbf{x})\,d\mathbf{x}$ (monotonicity).
6. Suppose $D = D_1 \cup D_2$ and the measure of $D_1 \cap D_2$ is zero. Then

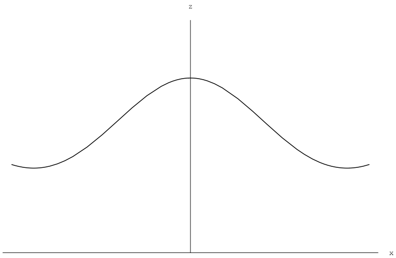
$$\int_D f(\mathbf{x})\,d\mathbf{x} = \int_{D_1} f(\mathbf{x})\,d\mathbf{x} + \int_{D_2} f(\mathbf{x})\,d\mathbf{x}.$$

These properties hold given that the relevant integrals exist.

Intuition behind iterated integrals

Let $D \subset \mathbb{R}^2$ be a rectangular region, and let $f : D \to \mathbb{R}$.





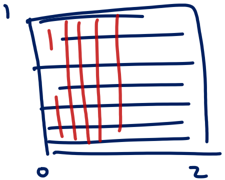
If we slice up the graph in one direction, so that each slice is the graph of a univariate function, and evaluate the integrals of these functions, these values give us a function that we can then integrate in the other variable and hopefully obtain the value of the integral over the region D.

A motivating example

Example 11.5

Let $D = [0,2] \times [0,1]$. Calculate

$$\iint_D 4 - x - y \, dA$$



firstly by slicing parallel to the xz-plane (x integral first) and secondly slicing parallel to the yz-plane (y integral first).

With the x-integral first:

$$\int_0^1 \int_0^2 4 - x - y \, dx \, dy = \int_0^1 \left[x(4-y) - \tfrac{1}{2}x^2 \right]_{x=0}^{2} dy$$

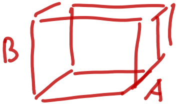
$$= \int_0^1 \big(2(4-y) - 2\big)\, dy = \int_0^1 6 - 2y \, dy = 6y - y^2 \Big|_{y=0}^{1} = 6 - 1 = 5$$

With the y-integral first:

$$\int_0^2 \int_0^1 4 - x - y \, dy \, dx =$$

$= 5$

General version of Fubini's theorem



Theorem 11.6 (Fubini, general version)

Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ be rectangular regions. Let $f : A \times B \to \mathbb{R}$ be integrable on $A \times B$. Then the functions

for fixed $\underset{\sim}{x}$

$$\left.\begin{aligned} L(\mathbf{x}) &= \sup\{\underline{S}_{\mathcal{P}_B}(\overbrace{f(\mathbf{x},\cdot)}^{\text{functions } g(\underset{\sim}{y}) = f(\underset{\sim}{x},\underset{\sim}{y})}) \mid \mathcal{P}_B \text{ partitions } B\}, \\ U(\mathbf{x}) &= \inf\{\overline{S}_{\mathcal{P}_B}(f(\mathbf{x},\cdot) \mid \mathcal{P}_B \text{ partitions } B\}, \end{aligned}\right\}$$

In a 'good case'

$$L(\underset{\sim}{x}) = U(\underset{\sim}{x}) = \int_B f(\underset{\sim}{x},\underset{\sim}{y})\, d\underset{\sim}{y}$$

are integrable on A and

$$\int_{A \times B} f = \int_A L = \int_A U.$$

Let $f : A \times B \to \mathbb{R}$. Fix $\underset{\sim}{x} \in A$, define $g(\underset{\sim}{y}) = f(\underset{\sim}{x},\underset{\sim}{y})$

$f(\underset{\sim}{x}, \cdot)$

Bonus: an example where $L(\mathbf{x}) \neq U(\mathbf{x})$

Example 11.7

Let $f : [0,1] \times [0,1] \to \mathbb{R}$ be defined as

$$f(x,y) = \begin{cases} 1 - 1/q, & \text{if } x, y \in \mathbb{Q}, \text{ and } x = p/q \text{ in the lowest terms,} \\ 1, & \text{otherwise} \end{cases}$$

Continuous version of Fubini's theorem

Theorem 11.8 (Fubini, continuous version)

Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ be rectangular regions. Let $f : A \times B \to \mathbb{R}$ be continuous on $A \times B$. Then

$$\int_{A\times B} f = \int_A \left(\int_B f(\mathbf{x}, \mathbf{y})\, d\mathbf{y} \right)\, d\mathbf{x} = \int_B \left(\int_A f(\mathbf{x}, \mathbf{y})\, d\mathbf{x} \right)\, d\mathbf{y}$$

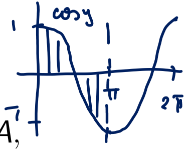
The integrals on the right-hand side are called *iterated* integrals.

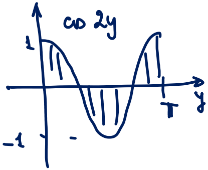
We usually apply Fubini's theorem in this form to sufficiently nice and possibly discontinuous functions, if necessary omitting the set of points of measure zero where $U(\mathbf{x}) \neq L(\mathbf{x})$.

Example 11.9

Use Fubini's theorem to calculate

$$I = \iint_D y\sin(xy)\,dA,$$





$$\left(-\frac{1}{y}\cos xy\right)'_x = \sin xy$$

where $D = [1,2]\times[0,\pi]$.

The two ways of calculating this integral are

$$\int_1^2\int_0^\pi y\sin(xy)\,dy\,dx \quad \text{and} \quad \int_0^\pi\int_1^2 y\sin(xy)\,dx\,dy.$$

The first one involves integration by parts, but not the second one.

$$I = \int_0^\pi \underbrace{\int_1^2 y\sin xy\,dx}\,dy = \int_0^\pi y\cdot\left(-\frac{1}{y}\right)\cos xy\Big|_{x=1}^{x=2} dy = -\int_0^\pi \cos 2y - \cos y\,dy$$
$$= 0$$

Example 11.10

Evaluate the triple integral

$$\iiint_R (xy^2 + z)\, dx\, dy\, dz$$

over the rectangular region R defined by $\boxed{1 \le x \le 2}$, $\underbrace{2 \le y \le 4}$, $\underbrace{0 \le z \le 1}$.

$$\int_1^2 \int_2^4 \boxed{\int_0^1 (xy^2 + z)\, dz}\, dy\, dx = \int_1^2 \int_2^4 zxy^2 + \frac{1}{2}z^2 \Big|_0^1\, dy\, dx$$

$$= \int_1^2 \int_2^4 x\underline{y^2} + \frac{1}{2}\, dy\, dx = \int_1^2 \left(\frac{1}{3}y^3x + \frac{1}{2}y\right)\Big|_{y=2}^{y=4} dx = \int_1^2 \frac{1}{3}x\left[4^3 - 2^3\right] + \frac{1}{2}\cdot 2\, dx$$

$$= \int_1^2 \left(\frac{64-8}{3}x + 1\right) dx = \int_1^2 \left(\frac{56}{3}x + 1\right) dx = \frac{28}{3}x^2 + x\Big|_1^2 = \frac{28}{3}\left[4-1\right] = \underline{29}$$

Theorem 11.11 (Corollary of Fubini's theorem)

If $f(\mathbf{x},\mathbf{y}) = g(\mathbf{x})\,h(\mathbf{y})$ and A, B are some rectangular regions (possibly line segments), the function g is continuous on A and the function h is continuous on B, then

$$\iint_{A\times B} f(\mathbf{x},\mathbf{y})\,d\mathbf{x}\,d\mathbf{y} = \left(\int_A g(\mathbf{x})\,d\mathbf{x}\right)\cdot\left(\int_B h(\mathbf{y})\,d\mathbf{y}\right).$$

doesn't depend on $\mathbf{x}$

$$\iint_{A\times B} f(\mathbf{x},\mathbf{y})\,d\mathbf{x}\,d\mathbf{y} = \int_A\left(\int_B g(\mathbf{x})\cdot h(\mathbf{y})\,d\mathbf{y}\right)d\mathbf{x} = \int_A g(\mathbf{x})\left(\int_B h(\mathbf{y})\,d\mathbf{y}\right)d\mathbf{x}$$

doesn't depend on $\mathbf{y}$

$$= \left(\int_B h(\mathbf{y})\,d\mathbf{y}\right)\cdot\left(\int_A g(\mathbf{x})\,d\mathbf{x}\right) \quad \checkmark$$

Example 11.12

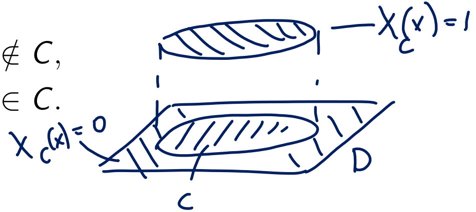
Calculate $I = \iint_D x^2 y^3 \, dx \, dy$ where $D = [0,1] \times [0,1]$.

By Theorem 11.11. $I = \int_0^1 x^2 \, dx \cdot \int_0^1 y^3 \, dy = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$

Integration over arbitrary regions

For $C \subset \mathbb{R}^n$ its characteristic function is defined as

$$\chi_C(x) = \begin{cases} 0, & \text{if } x \notin C, \\ 1, & \text{if } x \in C. \end{cases}$$



Theorem 11.13

Let D be a rectangular region, and $C \subseteq D$. The function $\chi_C : D \to \mathbb{R}$ is integrable if and only if the boundary of C has measure 0.

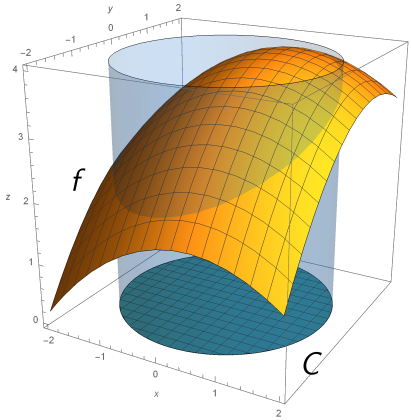
If $C \subset D$ for some rectangular region $D \subset \mathbb{R}^n$, then we define the integral of f over C as

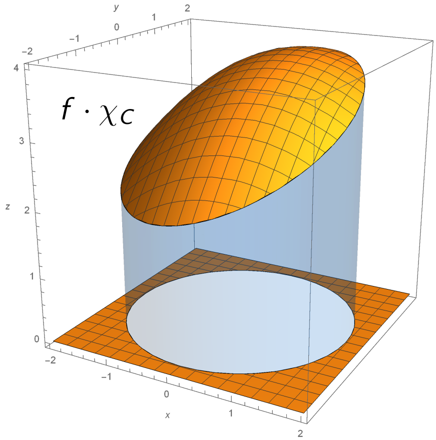
$$\int_C f = \int_D f \cdot \chi_C.$$

Integration over arbitrary regions

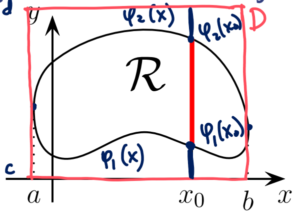
If $C \subset D$ for some rectangular region $D \subset \mathbb{R}^n$, then we define the integral of f over C as

$$\int_C f = \int_D f \cdot \chi_C.$$





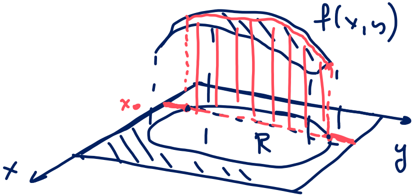
Integration over arbitrary regions



Suppose we are integrating f over $\mathcal{R}$ and we start with $x = x_0$ constant, as shown.
When we are evaluating the integral of f restricted to $x = x_0$, the limits of integration depend on x_0.

If $\mathcal{R} = \{(x, y) : a \le x \le b,\ \phi_1(x) \le y \le \phi_2(x)\}$ for two functions ϕ_1 and ϕ_2, then the integral over the slice shown at $x = x_0$ is

$$\int_c^d \underbrace{f(x_0,y)\cdot \chi(x_0,y)}\,dy = \int_{\phi_1(x_0)}^{\phi_2(x_0)} f(x_0, y)\, dy,$$



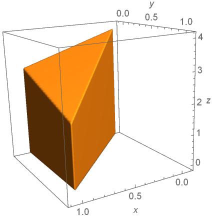
and the double integral over $\mathcal{R}$ is

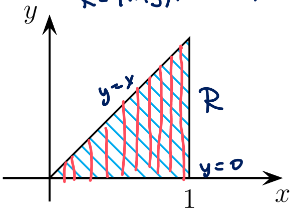
$$\iint_{\mathcal{R}} f\, dx\, dy = \int_a^b \left[\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy \right] dx.$$

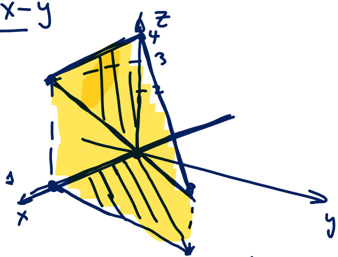
Example 11.14

Find the volume of the prism whose base is given by the triangle in the xy-plane bounded by the x-axis and the lines $y = x$ and $x = 1$ and whose top surface lies in the plane $x + y + z = 4$.

$R = \{(x,y) \mid 0 \le x \le 1,\ 0 \le y \le x\}$ $z = 4 - x - y$



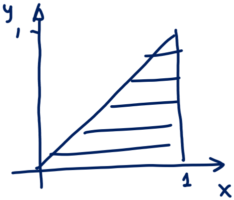


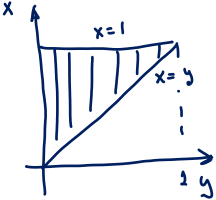


$$\iint_R 4-x-y\,dx\,dy = \int_0^1\int_0^x 4-x-y\,dy\,dx = \int_0^1 \left[(4-x)y - \tfrac{1}{2}y^2\right]_{y=0}^{y=x} dx$$

$$= \int_0^1 (4-x)x - \tfrac{1}{2}x^2\,dx = \int_0^1 4x - x^2 - \tfrac{1}{2}x^2\,dx = \int_0^1 4x - \tfrac{3}{2}x^2\,dx = 4\cdot\tfrac{1}{2}x^2 - \tfrac{3}{2}\cdot\tfrac{1}{3}x^3\Big|_{x=0}^{x=1}$$

$$= 2 - \tfrac{1}{2} = \tfrac{3}{2}$$





$$\int_0^1 \int_y^1 (4-x-y)\,dx\,dy = \dots\dots = \frac{3}{2}$$

Note about changing the limits

Notice that changing the order of integration gives

$$\int_0^1 \int_y^1 (4 - x - y)\, dx\, dy = \int_0^1 \int_0^x (4 - x - y)\; dy\, dx\,.$$

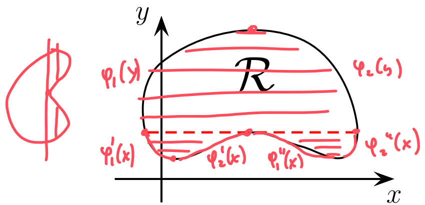
Unlike in the case of a rectangular region, we cannot just swap the two integrals around: we also need to recalculate the limits carefully.

Never do this:

$$\int_0^1 \int_0^x f(x,y)\, dy\, dx = \int_0^x \int_0^1 f(x,y)\, dx\, dy$$

$$\int_0^1 \int_0^y f(x,y)\, dx\, dy$$

We may encounter a problem when slicing a region in a particular way, for instance, holding y constant.



For the part of $\mathcal{R}$ below the dashed line, we cannot describe the x coordinate of the boundary using two functions of the y value of the slice, as such a slice cuts the boundary in four points, not two, and hence consists of two intervals, not one.

In this case we would have to either

- integrate with respect to y first (that is, hold x constant), or
- split $\mathcal{R}$ into three regions: a region above the dashed line and two regions below the line.

Example 11.15

Let I be the integral

$R = R_1 \cup R_2$, $R_1 = \{-5 \le y \le 0,\ -y \le x \le 5\}$

$R_2 = \{0 \le y \le 25,\ \underbrace{\sqrt{y} \le x}_{} \le 5\}$ $\quad y \le x^2,\ x \ge 0,\ y \ge 0$

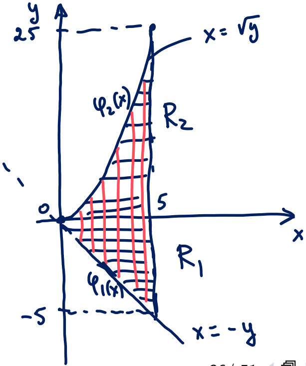
$$I = \underbrace{\overbrace{\int_{-5}^{0} \int_{-y}^{5}}^{R_1} \overbrace{(x+2y)}^{f(x,y)}\, dx\, dy}_{I_1} + \underbrace{\overbrace{\int_{0}^{25} \int_{\sqrt{y}}^{5}}^{R_2} \overbrace{(x+2y)}^{f(x,y)}\, dx\, dy}_{I_2}.$$

Sketch the region of integration, change the order of integration and hence calculate the value of I.

$R = \{0 \le x \le 5,\ -x \le y \le x^2\}$

$$I = \int_0^5 \underbrace{\boxed{\int_{-x}^{x^2} (x+2y)\, dy}}_{g(x)}\, dx =$$

$$g(x) = \int_{-x}^{x^2} x + 2y\, dy = \left. xy + 2 \cdot \tfrac{1}{2} y^2 \right|_{y=-x}^{x^2} = x(x^2+x) + x^4 - x^2 =$$



$\circled{=}$ $x^3 + \cancel{x^2} + x^4 - \cancel{x^2} = x^4 + x^3$

$$I = \int_0^5 g(x)\,dx = \int_0^5 x^4 + x^3\,dx = \frac{1}{5}x^5 + \frac{1}{4}x^4 \Big|_{x=0}^{x=5} = \frac{5^5}{4}$$

x-simple and y-simple regions

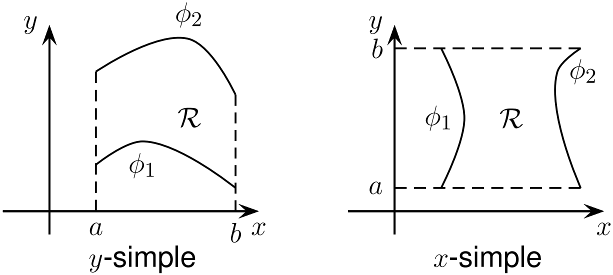
If the region $\mathcal{R}$ can be represented as

$$\mathcal{R} = \{(x, y) \,|\, a \le x \le b, \phi_1(x) \le y \le \phi_2(x)\},$$

this region is called y-**simple**. Likewise, if

$$\mathcal{R} = \{(x, y) \,|\, a \le y \le b, \phi_1(y) \le x \le \phi_2(y)\},$$

then this region is called x-**simple**.



Example 11.16

Let $\mathcal{R}$ be the region shown below. Find $I = \iint_{\mathcal{R}} xy\, dA$.

$2-x = x^2 \Rightarrow x^2+x-2 = (x-1)(x+2) = 0$

$x = 1$ or $x = -2$

① y- simple

② x- simple

① $R = \{(x,y) \mid -2 \le x \le 1,\ \varphi_1(x) \le y \le \varphi_2(x)\}$

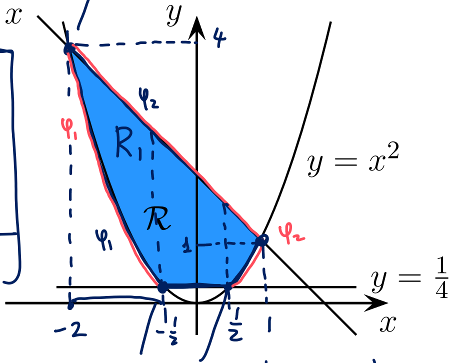
$= R_1 \cup R_2 \cup R_3$

$R_1 = \{(x,y) \mid -2 \le x \le -\frac{1}{2},\ x^2 \le y \le 2-x\}$

$R_2 = \{(x,y) \mid -\frac{1}{2} \le x \le \frac{1}{2},\ \frac{1}{4} \le y \le 2-x\}$

$R_3 = \{(x,y) \mid \frac{1}{2} \le x \le 1,\ x^2 \le y \le 2-x\}$

$$\int_{-2}^{1}\int_{x^2}^{2-x} xy\,dy\,dx - \int_{-\frac{1}{2}}^{\frac{1}{2}}\int_{x^2}^{\frac{1}{4}} xy\,dy\,dx$$



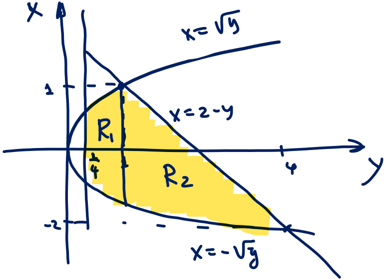
$x^2 = \frac{1}{4} \Rightarrow x = \frac{1}{2}$ or $x = -\frac{1}{2}$

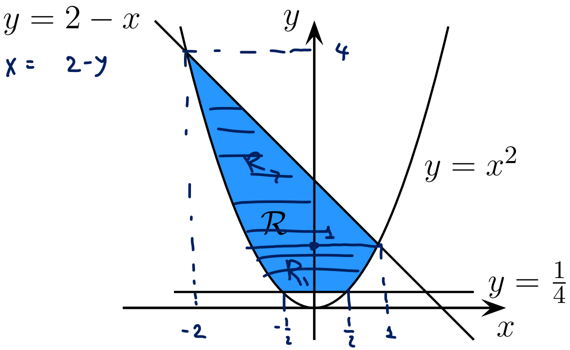
$$I = \int_{-2}^{-\frac{1}{2}}\int_{x^2}^{2-x} xy\,dy\,dx + \int_{-\frac{1}{2}}^{\frac{1}{2}}\int_{\frac{1}{4}}^{2-x} xy\,dy\,dx + \int_{\frac{1}{2}}^{1}\int_{x^2}^{2-x} xy\,dy\,dx$$

$= \quad = -\frac{45}{8}$

(2) x-simple

$$I = \int_{\frac{1}{4}}^{1} \int_{-\sqrt{y}}^{\sqrt{y}} xy \, dx \, dy + \int_{1}^{4} \int_{-\sqrt{y}}^{2-y} xy \, dx \, dy = \cdots = -\frac{45}{8}$$

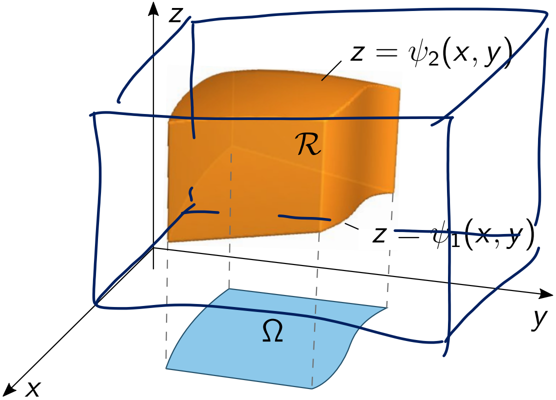




z-simple regions

If $\mathcal{R} \subset \mathbb{R}^3$ has the representation

$$\mathcal{R} = \{(x, y, z) : (x, y) \in \Omega,\ \psi_1(x, y) \leq z \leq \psi_2(x, y)\},$$



we call it z-**simple**, and the set Ω is called the xy-shadow of $\mathcal{R}$.

Triple integrals as iterated integrals

If $\mathcal{R}$ is a z-simple region with the shadow Ω, that is,

$$\mathcal{R} = \{(x,y,z) \,:\, (x,y) \in \Omega,\ \psi_1(x,y) \le z \le \psi_2(x,y)\},$$

then the triple integral of an integrable function f over $\mathcal{R}$ can be evaluated as the repeated integral

dy dx

dx dy

$$\iiint\limits_{\mathcal{R}} f(x,y,z)\,dx\,dy\,dz = \iint\limits_{\Omega} \left[\int_{\psi_1(x,y)}^{\psi_2(x,y)} f(x,y,z)\,dz\right] dA$$

$$= \iint\limits_{\Omega} \int_{\psi_1(x,y)}^{\psi_2(x,y)} f(x,y,z)\,dz\,dx\,dy$$

$$= \iint\limits_{\Omega} dx\,dy \int_{\psi_1(x,y)}^{\psi_2(x,y)} f(x,y,z)\,dz.$$

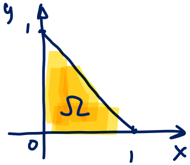
notation, not a product of two integrals!

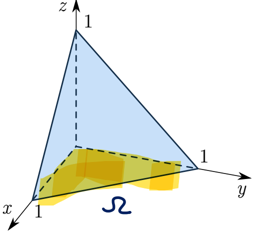
*The last expression is standard notation used for repeated integrals (it is **not** the product of two integrals).*

Example 11.17

Evaluate the triple integral

$$I = \iiint_{\mathcal{R}} \underbrace{x}_{f(x,y,z)}\, dV,$$





where $\mathcal{R}$ is the tetrahedron bounded by the xy, xz and yz planes and the plane $x + y + z = 1$. $\Rightarrow z = 1 - x - y$

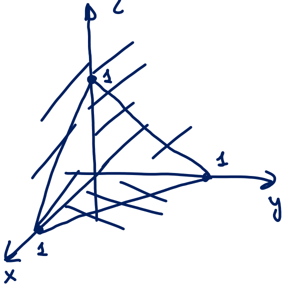
$$\Omega = \{(x,y) \mid x \geq 0, y \geq 0, x+y \leq 1\} = \{(x,y) \mid 0 \leq x \leq 1, 0 \leq 1-x\}$$

$$\psi_1(x,y) = 0, \quad \psi_2(x,y) = 1-x-y$$

$$0 \leq z \leq 1-x-y$$

$$\mathcal{R} = \{(x,y,z) \mid (x,y) \in \Omega, \psi_1(x,y) \leq z \leq \psi_2(x,y)\}$$

$$I = \iint_{\Omega} \left[\int_{\psi_1(x,y)}^{\psi_2(x,y)} x \, dz \right] dx\,dy = \,.$$



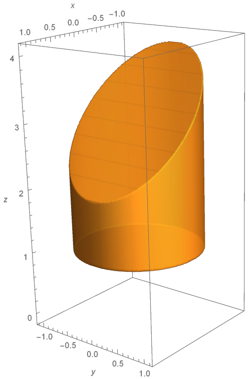
$$I = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} x \, dz = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} x \, dz \, dy \, dx$$

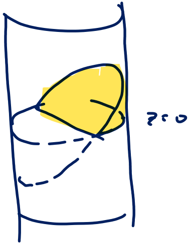
$$= \int_0^1 \int_0^{1-x} x z \Big|_{z=0}^{z=1-x-y} dy \, dx = \int_0^1 \int_0^{1-x} x(1-x-y) \, dy \, dx = \cdots = \frac{1}{12}$$

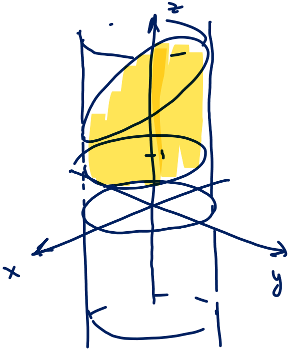
The hoof of Archimedes

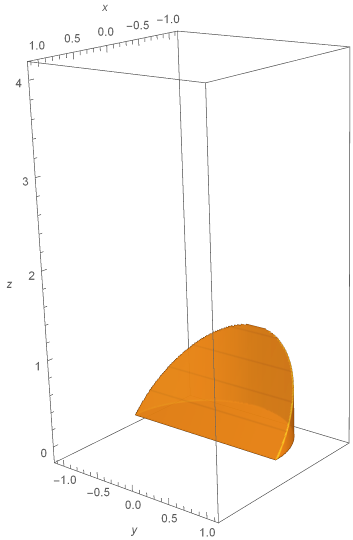
Example 11.18

Sketch the solid region bounded by $x^2 + y^2 = 1$, $z = 1$ and $z = 3 - x$. Calculate its volume. Do this for the hoof of Archimedes, which is the region bounded by $x^2 + y^2 = 1$, $z = 0$, $z = -x$.









x
1.0
0.5
0.0
−0.5
−1.0
z
4
3
2
1
0
y
−1.0
−0.5
0.0
0.5
1.0

Centroid and Centre of Mass

The **mass** of a solid occupying a three-dimensional bounded region $\mathcal{R}$, with the density $\delta(x, y, z)$ is

$$M = \iiint\limits_{\mathcal{R}} \delta(x, y, z)\, dV.$$

The coordinates of the **centre of mass** of this body are

$$\bar{x} = \frac{1}{M} \iiint\limits_{\mathcal{R}} x\delta(x, y, z)\, dV,$$

$$\bar{y} = \frac{1}{M} \iiint\limits_{\mathcal{R}} y\delta(x, y, z)\, dV,$$

$$\bar{z} = \frac{1}{M} \iiint\limits_{\mathcal{R}} z\delta(x, y, z)\, dV.$$

Centroid and Centre of Mass

Likewise, for a lamina (an object that can be considered flat) occupying a plane region $\mathcal{R}$ with the two-dimensional density given by $\delta(x, y)$, its total mass is $M = \iint_{\mathcal{R}} \delta(x, y)\, dA$, and the coordinates of the **centre of mass** of this lamina are

$$\bar{x} = \frac{1}{M} \iint_{\mathcal{R}} x\delta(x, y)\, dA, \qquad \bar{y} = \frac{1}{M} \iint_{\mathcal{R}} y\delta(x, y)\, dA.$$

The **centroid** of a region is the centre of mass of the region assuming uniform density. If we let $\delta(x, y, z) \equiv 1$, then $M = \mathrm{Vol}\,(R)$, and we have the coordinates of the centroid as

$$\bar{x} = \frac{1}{\mathrm{Vol}\,(\mathcal{R})} \iint_{\mathcal{R}} x\, dA, \qquad \bar{y} = \frac{1}{\mathrm{Vol}\,(\mathcal{R})} \iint_{\mathcal{R}} y\, dA,$$

with similar expressions for the three-dimensional regions.

Example 11.19

Find the centroid of the region enclosed by the three semi-circles below.